INCH-POUND

MIL-R-39016/39E
4 August 1994
SUPERSEDING
MIL-R-39016/39D
21 April 1989

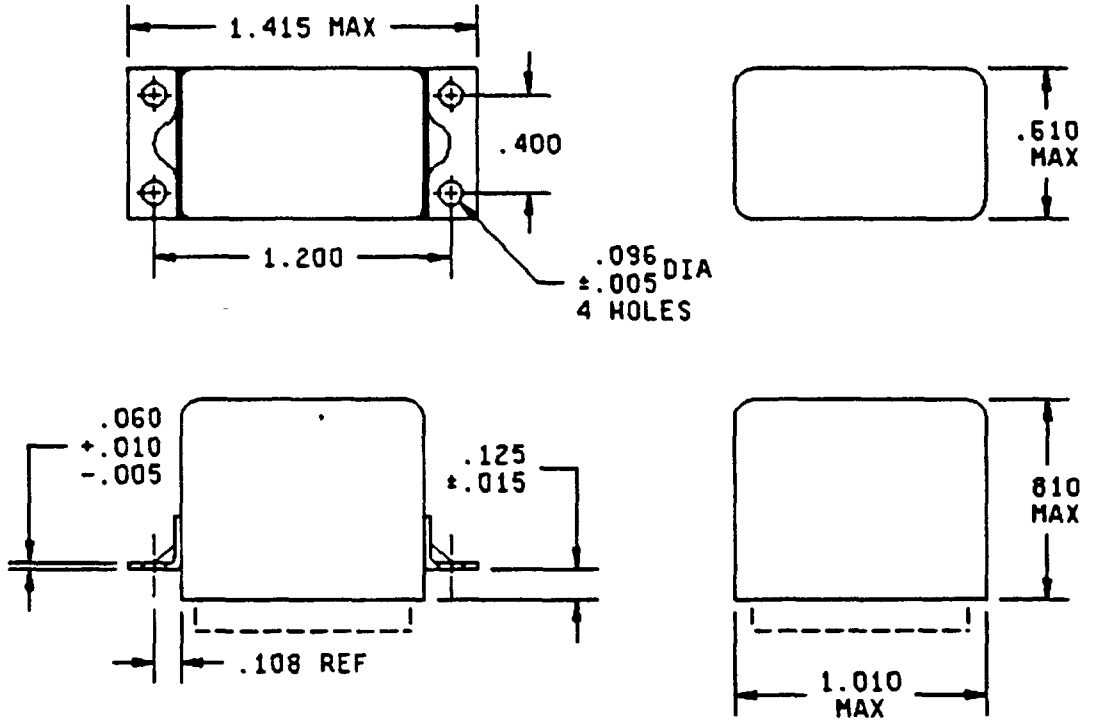
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, HERMETICALLY SEALED, 4PDT, LOW-LEVEL TO 3 AMPERES (400 MILLIWATTS)

Inactive for new design after 4 August 1994.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



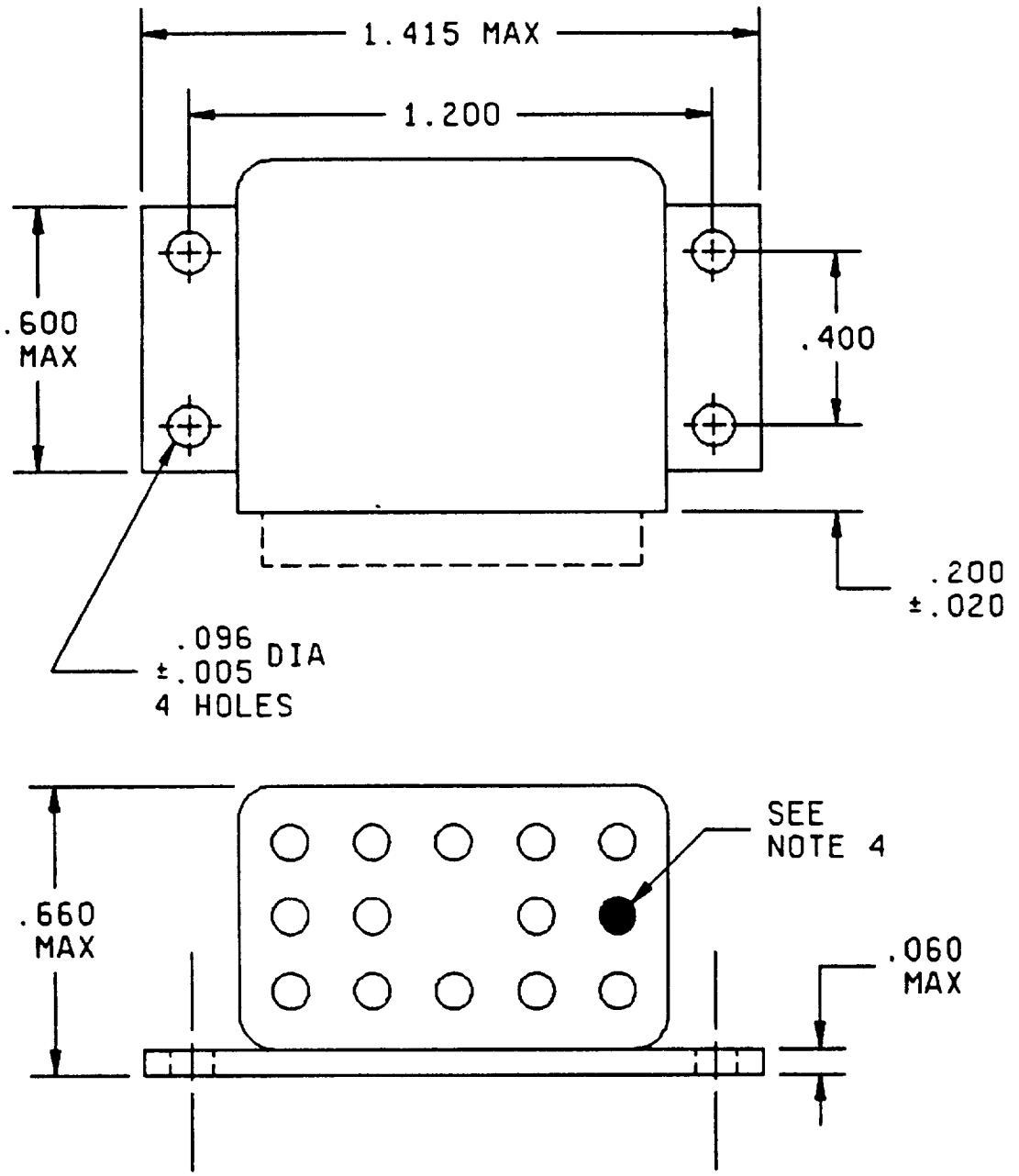
RAISED VERTICAL FLANGE MOUNT (RVFM)

NO MOUNT (NM)

(E) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

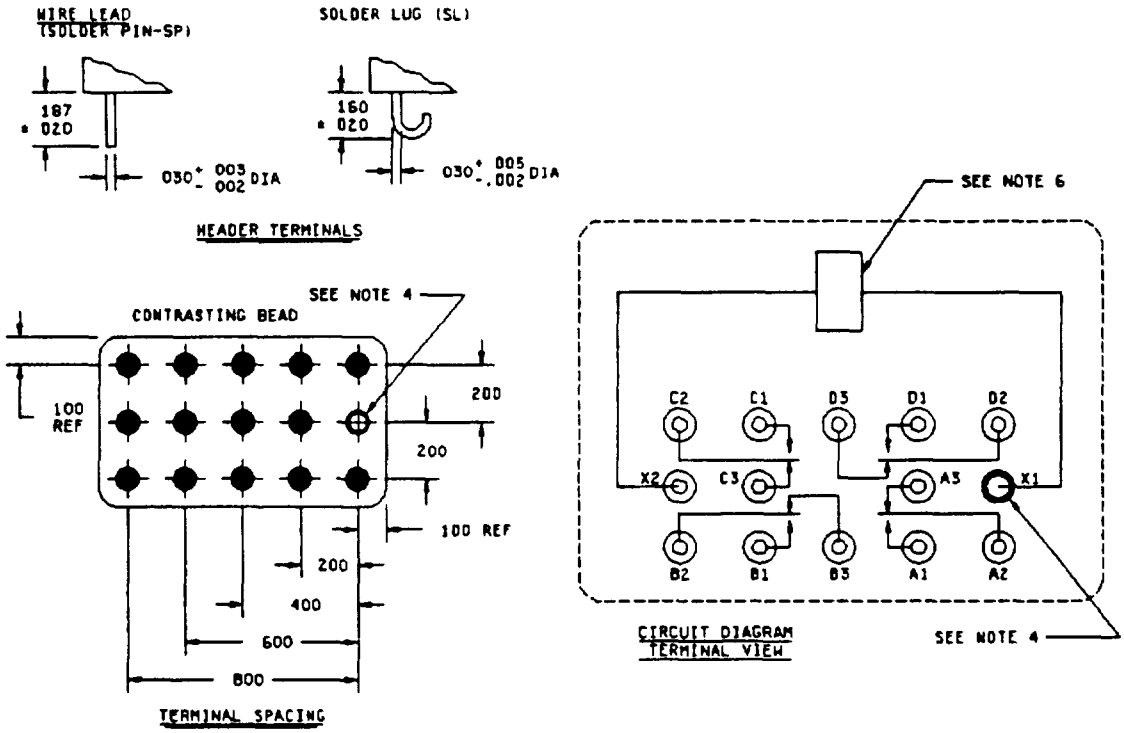
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HORIZONTAL FLANGE MOUNT (HFM)

FIGURE 1 Dimensions and configuration.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.060	1.52	.187	4.75	.800	20.32
.003	0.08	.096	2.44	.200	5.08	.810	25.57
.005	0.13	.100	2.54	.400	10.16	1.010	25.65
.010	0.25	.108	2.74	.600	15.24	1.200	30.38
.020	0.51	.125	3.18	.610	15.49	1.400	35.56
.030	0.76	.160	4.06	.660	16.76		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm) for three place decimals.
4. Indicated terminal shall be identified with a contrasting bead.
5. Terminal locating dimensions shown are applicable to all type mounts with a tolerance ±.005 (0.13 mm).
6. Coil symbol optional in accordance with MIL-STD-1285.
7. All terminals shall be marked on the circuit diagram.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded)

Resistive. 3.0 amperes at 28 V dc, 0.300 ampere at 115 V ac, 60 Hz and 400 Hz.

Resistive (relay case not grounded): 0.75 ampere at 115 V ac, 60 Hz and 400 Hz.

Inductive: 1.0 ampere at 200 mH inductive at 28 V dc (100,000 cycles).

Lamp: 0.200 ampere at 28 V dc

Low level: 10 µA to 50 µA at 10 mV to 50 mV dc or peak ac.

Intermediate current: 50,000 cycles.

Contact resistance or voltage drop

Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.100 ohm maximum.

Low level:

During life: 33 ohms maximum

After life 0.150 ohm maximum.

Intermediate current:

During intermediate current: 1 ohm maximum.

After intermediate current: 0.300 ohm maximum.

Contact bounce: 2 0 ms maximum.

Contact stabilization time: 2.5 ms maximum

Overload (high level only).

Resistive: Two times rated, except grounded case ac.

COIL DATA: (See table I.)

Operate time 8 ms maximum over temperature range with rated coil voltage

Release time· 8 ms maximum over temperature range from rated coil voltage.

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TABLE I. Dash numbers and characteristics. 1/

Dash no.	Mounting style (see figure 1)	Terminal code	Coil voltage (V dc)		At +25°C				Over temperature range		
			Rated 2/	Max	Coil resistance ohms ±10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
-001	RVFM	SL	26.5	32.0	500	13.5	7.0	2.0	18.0	11.5	1.0
-002	RVFM	SP	26.5	32.0	500	13.5	7.0	2.0	18.0	11.5	1 0
-003	NM	SL	26.5	32.0	500	13.5	7.0	2.0	18.0	11.5	1.0
-004	NM	SP	26.5	32.0	500	13.5	7.0	2.0	18.0	11.5	1.0
-005	HFM	SL	26.5	32.0	500	13.5	7.0	2.0	18.0	11.5	1.0
-006	HFM	SP	26.5	32.0	500	13.5	7.0	2.0	18.0	11.5	1 0
-007	RVFM	SL	12.0	16.5	117	6.6	3.4	1.0	8.8	5.65	0.48
-008	RVFM	SP	12.0	16.5	117	6.6	3.4	1.0	8.8	5.65	0.48
-009	NM	SL	12.0	16.5	117	6.6	3.4	1.0	8.8	5.65	0.48
-010	NM	SP	12.0	16.5	117	6.6	3.4	1.0	8.8	5.65	0.48
-011	HFM	SL	12.0	16.5	117	6.6	3.4	1.0	8.8	5.65	0.48
-012	HFM	SP	12.0	16.5	117	6.6	3.4	1.0	8.8	5.65	0.48
-013	RVFM	SL	6.0	8.5	32	3.4	1.7	0.5	4.55	2.9	0.24
-014	RVFM	SP	6.0	8.5	32	3.4	1.7	0.5	4.55	2.9	0.24
-015	NM	SL	6.0	8.5	32	3.4	1.7	0.5	4.55	2.9	0.24
-016	NM	SP	6.0	8.5	32	3.4	1.7	0.5	4.55	2.9	0.24
-017	HFM	SL	6.0	8.5	32	3.4	1.7	0.5	4.55	2.9	0.24
-018	HFM	SP	6.0	8.5	32	3.4	1.7	0.5	4.55	2.9	0.24

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

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ELECTRICAL DATA.

Insulation resistance: 1,000 megohms minimum, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

Sea level:

Between case, frame or enclosure, and all contacts in the energized and deenergized positions	1,000 V rms (60 Hz).
Between case, frame or enclosure, and coils	500 V rms (60 Hz).
Between all contacts and coils	1,000 V rms (60 Hz).
Between open contacts in the energized and deenergized positions	500 V rms (60 Hz).
Between contact poles	1,000 V rms (60 Hz).

Altitude:

Between all terminals to case	350 V rms (60 Hz).

ENVIRONMENTAL DATA:

Temperature range. -65°C to +125°C.

Vibration (sinusoidal): Method 204, MIL-STD-202. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts

Vibration (random): Method 214, MIL-STD-202, test condition IG. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Shock (specified pulse): Method 213, MIL-STD-202, test condition A, 50 g's (see figure 1, HFM); test condition B, 75 g's (see figure 1, RVFM and NM). Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Resistance to soldering heat: Applicable.

PHYSICAL DATA:

Terminals: See figure 1 and table I.

Terminal strength: 3 pounds ±0.2 pound (pull).

Solderability: Applicable.

Dimensions and configurations: See figure 1.

Weight: 30 grams (1.06 ounce) maximum.

Identification marking (full): Applicable.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles.

Low level: 100,000 cycles.

PART OR IDENTIFYING NUMBER (PIN): M39016/39- (dash number from table I).

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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
50 units plus 1 open unit. One failure allowed.	M39016/39-001	50 units plus 1 open unit. One failure allowed.
Qualification inspection as applicable.		Qualification inspection as applicable.
	M39016/39-018	Two units, qualification, group II, also shock, vibration, acceleration, terminal strength and seal.

1/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection testing, the relay manufacturer shall preselect the sampling plan.

SUPERSESSION DATA: See table III

TABLE III. Supersession data.

Superseded PIN M5757/12-	New PIN M39016/39-
001	001
003	002
005	003
007	004
009	005
011	006
013	007
015	008
017	009
019	010
021	011
023	012
025	013
027	014
029	015
031	016
033	017
035	018

QUALITY ASSURANCE PROVISIONS:

Groups B and C not required. Group A required. The qualifying activity shall be notified of any design and/or construction changes and shall impose additional testing requirements as necessary.

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CONCLUDING MATERIAL

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - SH

Preparing activity:
DLA - ES

(Project 5945-0947-02)

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